

Ensemble Intrusion Detection Based on Heterogeneous Data Augmentation and Knowledge Distillation

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Abstract—The number and complexity of network attacks and intrusion events are constantly increasing, timely detection of abnormal intrusion behavior is an important challenge in the field of network security. To this end, this article proposes an ensemble intrusion detection model based on heterogeneous data augmentation and mutual knowledge distillation (KD), KDEHDA. To enhance the diversity of data, the training set is divided into multiple data subsets using the bootstrap sampling method, and different data augmentation methods are used on each subset to obtain multiple balanced data subsets. To enhance the generalization ability of the model, different base CNN models are trained using balanced data subsets, and KD is used to transfer the weight information of CNN. The final result is the voting ensemble of each CNN model. Experimental results show that the proposed model is superior to the existing optimal model.

Index Terms—Data augmentation, Ensemble learning, Intrusion detection, Knowledge distillation.

I. INTRODUCTION

NETWORK attack techniques have become increasingly covert and complex. Intrusion detection, an effective active defense strategy, has garnered growing attention. Traditional intrusion detection models incorporating machine learning are gradually proving inadequate in managing this increasingly complex attack environment [1]. In recent years, deep learning models have been widely applied in intrusion detection. The

multilayered structure and high complexity of deep learning models enable them to capture abstract, nonlinear relationships in data—crucial for identifying complex attack patterns. Compared to traditional machine learning, deep learning more comprehensively analyzes the internal structure of data, enhancing its ability to detect hidden threats. Moreover, as hardware computing power improves and algorithm optimization advances, the efficiency and effectiveness of deep learning continue to improve, further advancing its role in intrusion detection [2].

Despite deep learning’s impressive performance in intrusion detection, its complexity and high demand for computational resources present substantial challenges. Deep learning models typically involve numerous parameters and hierarchical structures, leading to substantial time and resource consumption and, consequently, high computational costs. In practical applications, this computational cost becomes a significant barrier to deploying deep learning models [3]. For resource-constrained edge devices, such as IoT and mobile devices, large deep learning models are challenging to deploy directly.

Knowledge distillation (KD), a widely recognized model compression technique, has been extensively applied in intrusion detection [4]. It compresses models by transferring knowledge from a large, high-precision model, or “teacher model,” to a smaller, lightweight model, known as the “student model.” This process enables the student model to substantially reduce computing resource requirements while maintaining a degree of accuracy. This allows for the deployment of compact, distilled deep learning models that can perform efficient intrusion detection, even on resource-constrained edge devices. However, traditional KD methods face two main issues [5]. First, the pre-training of teacher networks is often resource-intensive. Achieving a high-precision teacher model usually requires multiple adjustments and optimizations, making the process complex and time-consuming. Second, architectural differences between the teacher and student models may impact the effectiveness of KD. Research shows that if the disparity between teacher and student models is too great, it may not only hinder effective learning but could also degrade performance [6]. These challenges are also encountered when applying KD to build lightweight intrusion detection models.

We observed that ensemble learning enhances classification by constructing and combining multiple classifiers, effectively optimizing weaker models [7]. In addition, the complexity

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of convolutional neural networks escalates exponentially with added hidden layers. By integrating multiple shallow networks, we can construct a lightweight intrusion detection model with both reduced capacity and high accuracy. The main contributions are as follows.

- 1) We propose a novel KD method, convolutional knowledge distillation (CKD), enables the transfer of knowledge across different data types within the model, enhancing its adaptability.
- 2) We apply multiple data augmentation methods to process intrusion detection datasets, effectively increasing data diversity and improving model robustness in handling imbalanced datasets.
- 3) We develop a new deep ensemble learning framework by combining CKD with heterogeneous ensemble learning, to achieve improved classification performance in intrusion detection tasks.

The rest of this article are organized as follows. In Section II, we outline recent advancements in deep learning-based intrusion detection models, including some lightweight models. Section III discusses the rationale for introducing heterogeneous data augmentation and convolutional distillation and explains the research methodology. Section IV presents the experimental framework, includes ablation experiments to demonstrate the effectiveness of convolutional distillation, and provides the experimental results of our method alongside other comparative methods. Finally, Section V concludes this article.

II. RELATED WORK

In recent years, KD and ensemble learning have emerged as promising solutions for building efficient and accurate intrusion detection systems. KD allows for transferring knowledge from a complex teacher model to a simpler student model, thereby reducing model complexity while maintaining performance. For instance, Wang et al. [8] proposed a lightweight model guided by a teacher with a stable loss function for better convergence. WangX et al. [9] employed a spatiotemporal residual teacher network to guide a compact student detector through multi-scale KD. Most existing KD methods can be categorized into response-based, feature-based, and relation-based approaches. Response-based methods transfer softened logits, while feature-based methods align intermediate representations. More recently, relation-based distillation, such as relational knowledge distillation (RKD) [10], has gained attention for transferring the structural relationships among instances instead of direct feature values.

While these methods show effectiveness, they typically rely on a manually designed or pretrained teacher model, which introduces significant computational overhead and limits adaptability to new domains or datasets. Furthermore, most prior KD-based intrusion detection models do not incorporate diverse data views or tackle class imbalance simultaneously.

To overcome these limitations, ensemble-based approaches have been adopted to improve robustness and generalization. However, existing ensembles often lack intermodel communication during training, resulting in limited knowledge sharing [11].

In contrast, our proposed KDEHDA framework introduces a novel ensemble-based CKD mechanism, where multiple student models are trained collaboratively using heterogeneous data augmentations. The mutual distillation among peer models eliminates the need for a fixed teacher and enables adaptive knowledge transfer. In addition, by integrating various augmentation strategies tailored to the minority class, our method addresses data imbalance and overfitting simultaneously.

III. THE PROPOSED METHOD

This section outlines the proposed method, with Section III-A detailing the algorithm's motivation, Section III-B presenting the principle of the Bootstrap sampling method, Section III-C explaining the CKD, and Section III-D providing the specific process of KDEHDA.

A. Motivation

In the field of large-scale data applications, deep neural networks are increasingly supplanting traditional machine learning methods and have proven highly effective for complex tasks, such as intrusion detection. However, binary classification in intrusion detection frequently encounters severe class imbalance [12], prompting researchers to employ data augmentation to enhance dataset size by generating additional instances. Data augmentation may exhibit variable efficacy across data types, and a single method may be inadequate to capture the full diversity of the data, increasing the risk of overfitting or underfitting.

This study addresses these issues by proposing KDEHDA, a novel ensemble KD CNN framework. KDEHDA uses heterogeneous data augmentation, assigning distinct data subsets to each student model, processed using diverse augmentation techniques. This method diversifies training instances, thereby enhancing robustness and adaptability across different environments. KDEHDA eliminates reliance on a pretrained teacher model; instead, a mutual learning mechanism facilitates knowledge sharing by transferring convolutional kernel weights from the highest-performing student model to others, which fosters flexibility in knowledge transfer and improves model generalization without sacrificing computational efficiency—ideal for resource-constrained intrusion detection systems.

B. Bootstrap Sampling

The bootstrap sampling is a resampling method that incorporates replacement. Given a dataset containing m instances, each instance is randomly selected and added to the sampling set, then returned to the initial data set, allowing it to be selected again in subsequent draws. Following random m draws, a sampling set of size m is produced. Certain instances in the original data set may appear multiple times in the sampling set, while others may not be included at all. The probability of any sample not being selected during these m sampling operations is $(1 - \frac{1}{m})^m$, taking the limit we get.

The instances that remain unsampled after multiple iterations constitute the out-of-bag samples, which can be used as

validation sets to assess the classification performance of each subset

$$\lim_{m \rightarrow \infty} \left(1 - \frac{1}{m}\right)^m = \frac{1}{e} \approx 0.368. \quad (1)$$

C. Convolutional Knowledge Distillation

Traditional KD methods can be broadly categorized into output KD [13] and intermediate-layer KD [14], both of which typically require the student and teacher models to be trained on identical or highly similar datasets. This dependence limits their applicability in heterogeneous or domain-shifted scenarios.

To address this limitation, we propose a novel technique called CKD, which decouples the distillation process from the training data. Instead of transferring knowledge via output logits or feature maps, CKD directly transfers the convolutional kernels learned by the teacher network. This approach allows knowledge transfer between models trained on different datasets, enhancing flexibility and generalization in multisource or heterogeneous environments.

Assume that the convolution kernel $\mathbf{K}$ is a 3-D tensor with shape (M, H, W) , where M represents the number of output channels, in our experiments, we set the number of output channels to be the same, H and W denote the height and width of the kernel, respectively. $\mathbf{K}$ can be represented as a set of points within a 3-D matrix

$$\mathbf{K} = \left\{ K_{m,h,w} \mid m = 1, 2, \dots, M, \right. \\ \left. h = 1, 2, \dots, H, w = 1, 2, \dots, W \right\}. \quad (2)$$

Generally, convolution kernels typically have sizes of 3×3 or 5×5 , convolution kernels are employed to compute the distillation loss. When convolution kernel sizes are inconsistent, interpolation is used to resize the smaller kernel to match the dimensions of the larger kernel. This interpolation method applies bilinear interpolation, a technique that performs linear interpolation in two directions. In a 2-D plane, given a set of known points, bilinear interpolation estimates the value of an unknown point. Bilinear interpolation involves performing linear interpolation first in the horizontal direction, then in the vertical direction, interpolating across H and W (the height and width of the convolution kernel). For example, change the size (H, W) to (H', W')

$$K'(x', y') = \sum_{i=0}^1 \sum_{j=0}^1 \omega_{i,j} \cdot K(x_i, y_j) \quad (3)$$

where $K'(x', y')$ denotes the value of the interpolated convolution kernel at coordinate (x', y') , $K(x_i, y_j)$ is the value of the original convolution kernel at the four neighboring points (x_0, y_0) , (x_0, y_1) , (x_1, y_0) , and (x_1, y_1) , $\omega_{i,j}$ is interpolation weight

$$\omega_{i,j} = (1 - |x' - x_i|) \cdot (1 - |y' - y_j|). \quad (4)$$

Bilinear interpolation is well-suited for the continuous scaling of weight tensors, particularly for the smooth adjustment of convolution kernel weights.

When the convolution kernels of the teacher and student networks are of the same size, CKD employs the mean squared

error between the convolution kernels of the final convolution layer of the student and teacher models as the distillation loss

$$L_{MSE} = \frac{1}{MHW} \sum_{m=1}^M \sum_{h=1}^H \sum_{w=1}^W (K_{m,h,w}^t - K_{m,h,w}^s)^2 \quad (5)$$

where $K_{m,h,w}^t$ and $K_{m,h,w}^s$ denotes the weights of the last convolutional kernel of the student model and the teacher model at the coordinates (m, h, w) .

The key innovation of our method lies in its ability to perform KD directly on model parameters (convolution kernels), independent of the underlying data. This enables cross-domain and heterogeneous-model knowledge transfer, which is not feasible with traditional distillation methods. Our proposed CKD opens new avenues for collaborative learning and knowledge reuse in diverse data environments.

D. KDEHDA

The KDEHDA process comprises two main stages: Data preprocessing and model training, which are detailed in Sections III-D1 and III-D2.

1) *Data Preprocessing*: Initially, the intrusion detection dataset undergoes preprocessing to ensure compatibility with deep learning models. The dataset includes both numerical and nonnumerical features. Nonnumerical features are first transformed using one-hot encoding, categorical labels are processed via label encoding, and numerical features are scaled using min-max normalization. To conform to the input format required by 2D-CNN models, the 1-D feature vectors are reshaped into two-dimensional matrices.

To enable heterogeneous training environments and robust model generalization, following the common setting used in practical implementations, such as Scikit-learn [15]. Specifically, we draw five training subsets from the original dataset via bootstrapping using a sampling ratio of 77%. Each subset is independently sampled, and approximately 23% of the original instances are excluded from each sampling. These out-of-bag instances from all five rounds are then merged and deduplicated to construct a unified validation set. This ensures that the validation set remains diverse, comprehensive, and disjoint from the training subsets.

To promote diversity and address class imbalance within the training subsets, only the first training subset is left unmodified, serving as a baseline. The remaining four subsets are augmented using different oversampling strategies: SMOTE, ROS, ADASYN, and GAN-generated synthetic data. In each case, the minority class is oversampled to match the majority class, enhancing class balance and learning robustness. This process yields five distinct training subsets with varying characteristics, which are later used for ensemble learning.

2) *Model Training*: During training, multiple CNN models are trained on the balanced data subsets obtained through different augmentation. The first training round serves as an initialization phase. In subsequent rounds, each CNN model is evaluated at the start of each round using the validation set, with Macro-F1 as the evaluation metric. The CNN model achieving the highest Macro-F1 score is designated as the

Algorithm 1: Calculate Distillation Loss.

Input: Convolution kernel in the last layer of the student $\mathbf{K}^s$, convolution kernel in the last layer of the teacher $\mathbf{K}^t$, current epoch t , total epoch T , student output $\mathbf{q}_i^{\mathbf{T}(\mathbf{x})}$, real label $\mathbf{y}$

Output: Distillation loss L

```

1: Begin
2: for  $t = 1$  to  $T$  do
3:   while  $H^s \neq H^t$  or  $W^s \neq W^t$  do
4:      $\mathbf{K}(H', W') \leftarrow \mathbf{K}(H, W)$ 
5:   end while
6:    $L_{\text{soft}} \leftarrow (\mathbf{K}^s, \mathbf{K}^t)$ 
7:    $L_{\text{hard}} \leftarrow h(\mathbf{q}_i^{\mathbf{T}(\mathbf{x})}, \mathbf{y})$ 
8:    $\alpha(t), \beta(t) \leftarrow (t, T)$ 
9:    $L \leftarrow \alpha(t) * L_{\text{soft}} + \beta(t) * L_{\text{hard}}$ 
10:   $t \leftarrow t + 1$ 
11: end for
12: End

```

teacher model for that round, with the remaining models serving as student models. Macro $- F1$ represents the average $F1$ score across all categories

$$\text{Macro} - F1 = \frac{\sum_{i=1}^n F1_i}{n} \quad (6)$$

where $F1_i$ denotes the $F1$ of the category.

Macro $- F1$ is employed as the evaluation metric for selecting the teacher model, as it comprehensively assesses the model's classification performance across all classes. This study balances the dataset using a heterogeneous data augmentation algorithm to improve classification performance on minority classes. However, balancing the dataset with data augmentation introduces a risk of overfitting. By using Macro $- F1$, we maintain the model's recognition capability for the majority class, allowing for the selection of the optimal student as the teacher model to guide other student models.

During training, the loss function for the CNN teacher model is the cross-entropy loss, while the student model's loss function is a weighted combination of the cross-entropy loss and the convolutional distillation loss. The student model's loss function can be expressed as

$$\begin{aligned}
L &= \alpha(t) \cdot L_{\text{soft}} + \beta(t) \cdot L_{\text{hard}} \\
&= \alpha(t) \cdot \frac{1}{MHW} \sum_{m=1}^M \sum_{h=1}^H \sum_{w=1}^W (K_{m,h,w}^t - K_{m,h,w}^s)^2 \\
&\quad + \beta(t) \cdot h(\mathbf{q}_i^{\mathbf{T}(\mathbf{x})}, \mathbf{y}). \quad (7)
\end{aligned}$$

In KDEHDA, $\alpha + \beta = 1$, and we aim to avoid keeping the weights α and β fixed. We set β to follow a cosine decay function with respect to the epoch, gradually decreasing as the epoch progresses, while α increases correspondingly. In the early stages of training, each CNN model may not yet be proficient in the task. At this stage, setting a larger β facilitates the completion of basic tasks by each CNN model and improves performance

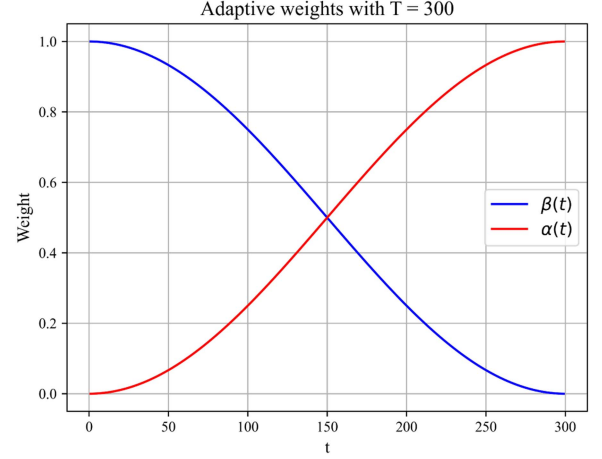


Fig. 1. Adaptive weights α and β .

on its respective training set. The later iterations correspond to the fine-tuning stage of KDEHDA. As training progresses, each CNN model approaches convergence. By gradually reducing the weight of the hard loss, CNN model can enhance its performance through the guidance of superior CNN models, thereby reducing the risk of overfitting the training data. α and β are represented as follows:

$$\beta(t) = \frac{1}{2} \left(1 + \cos \left(\frac{\pi t}{T} \right) \right), \alpha(t) = 1 - \beta(t). \quad (8)$$

When the total number of iterations T is 300, the curves of α and β changing with the number of iterations t are shown in Fig. 1. This design is inspired by the cosine learning rate schedule in SGDR [16], which has demonstrated effectiveness in improving convergence and reducing overfitting.

The procedure of the calculate distillation loss is described in the Algorithm 1. The final result is determined through hard voting among the CNN models, the process of KDEHDA is illustrated in Fig. 2.

IV. EXPERIMENT STUDY

To illustrate the effectiveness of the proposed method, four publicly available intrusion detection datasets are chosen for experiments. In addition, set of ablation experiment are devised to showcase the effectiveness of each component. Section IV-A outlines the experimental setup, Section IV-B details the ablation experiments, and Section IV-C provides the final experimental results and analysis.

A. Experimental Setting

1) *Experimental Environment:* The experiments were executed on a system operating the Windows 11 operating system. The hardware configuration included an Intel Core i5-12400F processor and 16 GB of RAM. The Python programming language version 3.7.0 was used for implementation, and the PyCharm integrated development environment (IDE) was employed for coding and project management. Based on the Pytorch framework, the GPU is a GeForce RTX2070 s 8 GB.

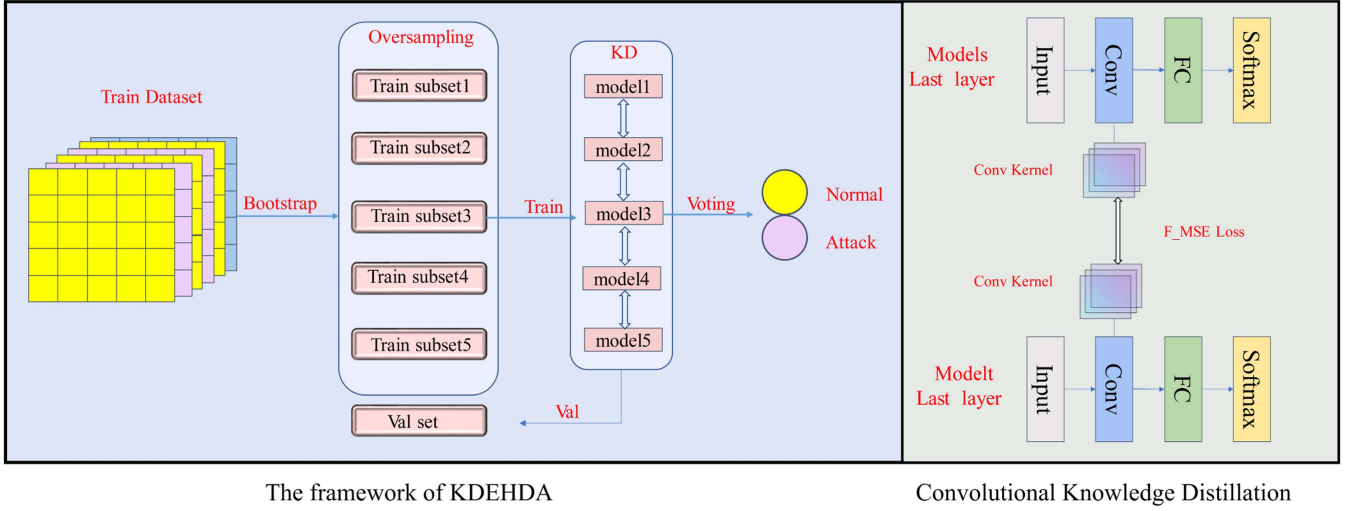


Fig. 2. KDEHDA flow chart.

TABLE I
DATASET CHARACTERISTICS AND INTRUSION STATISTICS

Dataset	Features	Training set			Testing set		
		Normal	Intrusions	Attack ratio	Normal	Intrusions	Attack ratio
KDDCUP99	41	97 278	396 743	80.30%	60 593	250 436	80.50%
UNSWNB15	42	37 000	45 332	55.10%	56 000	119 341	68.10%
CICIDS17	78	80 000	20 000	20%	720 000	180 000	20%
AAGM17	79	80 000	20 000	20%	80 000	20 000	20%
WEB-IDS23	82	80 000	20 000	20%	80 000	20 000	20%

2) **Datasets:** We collected four datasets from the University of New Brunswick website, namely KDDCUP99, UNSWNB15, CICIDS17, and AAGM17. KDDCUP99 and UNSWNB15 have provided divided training sets and validation sets. For CICIDS17 and AAGM17 datasets, we follow previous work [17] to divide the datasets to ensure fairness in comparative experiments. In addition, to enhance the real-world applicability of our evaluation and account for evolving threats, we incorporated the recently published WEB-IDS23 dataset [18], which contains up-to-date network traffic and attack scenarios. The dataset description is shown in Table I.

3) **Parameter Information:** We use three-layer CNN as the base model, with an initial 8 channels, 32 output channels in the final layer, dropout layer is applied to each layer to mitigate overfitting. In our previous study, we observed that when using CNN as the base model, the loss function consistently converged within approximately 200 epoch across the four datasets [19], so we set the epoch to 300. Additional parameters are provided in Table II.

4) **Evaluation Metrics:** The evaluation metrics of the intrusion detection model are the overall accuracy (OA) and F1. OA is the ratio of correct predictions in the prediction results.

TABLE II
PARAMETER INFORMATION

	CNN model
Batch size	{32, 64, 128}
Learning rate	{0.00001, 0.001}
Dropout	[0, 1]
Kernel size	[3, 5]
Epoch	300

F1 is the harmonic mean of precision and recall, where precision measures the ability of the intrusion detection system to identify only attacks, while recall can be considered as the ability of the system to discover all attacks.

B. Ablation Experiment

1) **Experiment on MNIST:** To demonstrate the effectiveness of the proposed CKD method in a controlled and interpretable setting, we conducted an ablation study using the MNIST dataset. Although MNIST is not a binary classification dataset, such as intrusion detection benchmarks, it is widely used in the deep learning community for evaluating novel training strategies due to its simplicity and well-understood characteristics. Using

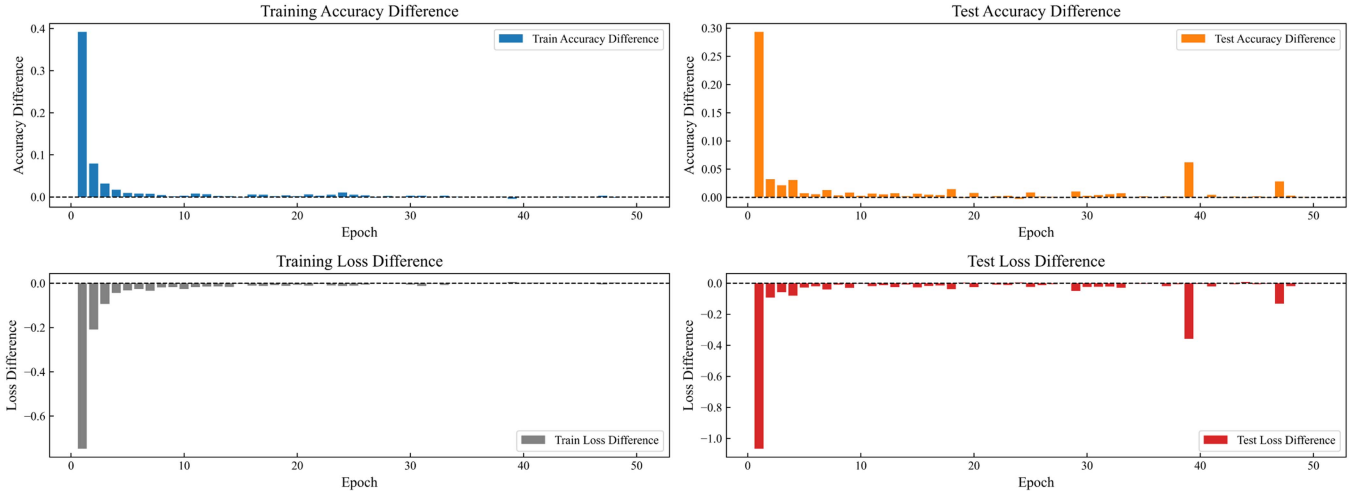


Fig. 3. Comparison of accuracy and loss on training and validation sets with and without KD using teacher model (3×3).

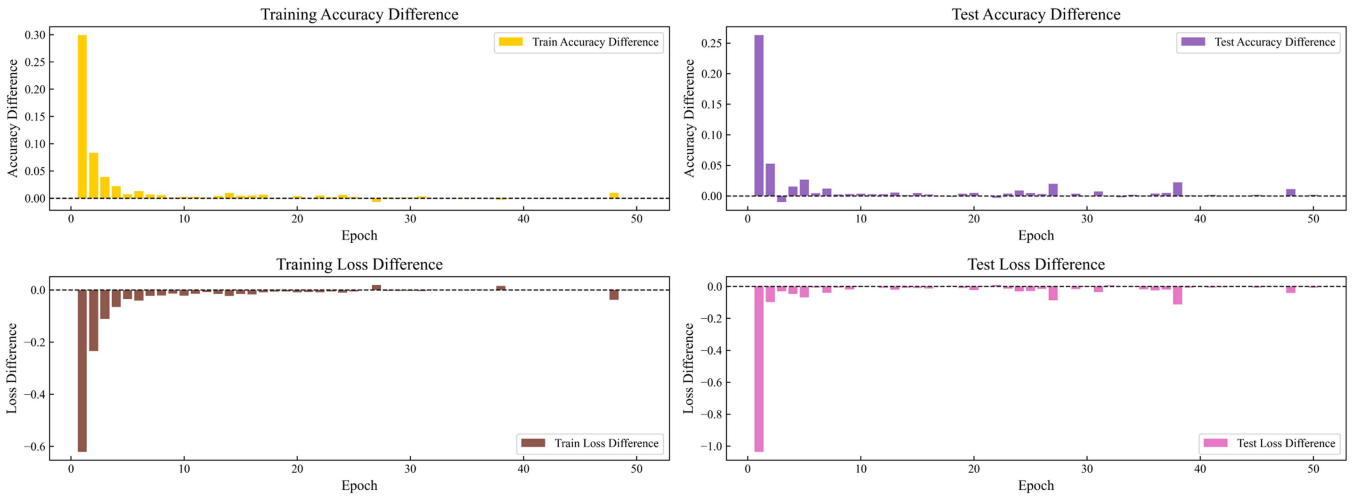


Fig. 4. Comparison of accuracy and loss on training and validation sets with and without KD using teacher model (5×5).

MNIST allows us to isolate and visualize the impact of CKD without the interference of task-specific complexities such as class imbalance or data noise.

In this experiment, two teacher models were pretrained using the full MNIST training set: one with 3×3 convolution kernels and the other with 5×5 kernels, representing different receptive field configurations. To simulate a low-resource scenario, we trained student models using only 10% of the original training data, thereby emphasizing the potential of KD to boost performance when labeled data is scarce.

The student models were trained with and without the proposed CKD, where the distillation loss used weighting factors $\alpha = \beta = 0.5$. Throughout training, we recorded accuracy and loss on both training and validation sets across epochs. Furthermore, the performance differences between student models with and without distillation are visualized in Figs. 3 and 4. As observed, the performance gap is more pronounced during the early training stages, where KD accelerates learning and guides the student toward more generalizable representations. After the 10th epoch, the difference narrows, which is expected as

both models gradually converge. However, the sporadic performance peaks throughout training demonstrate that KD provides a more stable optimization path, and overall, the distilled models achieve higher final accuracy and lower loss.

2) Experiment on AAGM17 and CICIDS17: To demonstrate the effectiveness of CKD, we use the CICIDS17 and AAGM17 datasets as examples. We set the number of iterations to 100. KD refers to CKD without a fixed teacher, while Null represents each independent CNN model, where the distillation loss is not computed. Only the original cross-entropy loss is used for back-propagation. The *OA* and *F1* scores for each CNN validation set are recorded in every training round. AAGM17 and CICIDS17 represent CNNs trained on data subsets without augmentation, while the remaining models are trained on balanced data subsets enhanced by various data augmentation techniques. The results are plotted as a line graph, as illustrated in the Figs. 5 and 6.

From the results in Fig. 5 it can be observed that on the AAGM dataset, the ordinate value corresponding to the GAN algorithm is the largest, indicating that the classification performance of the data subset enhanced by GAN outperforms that of other

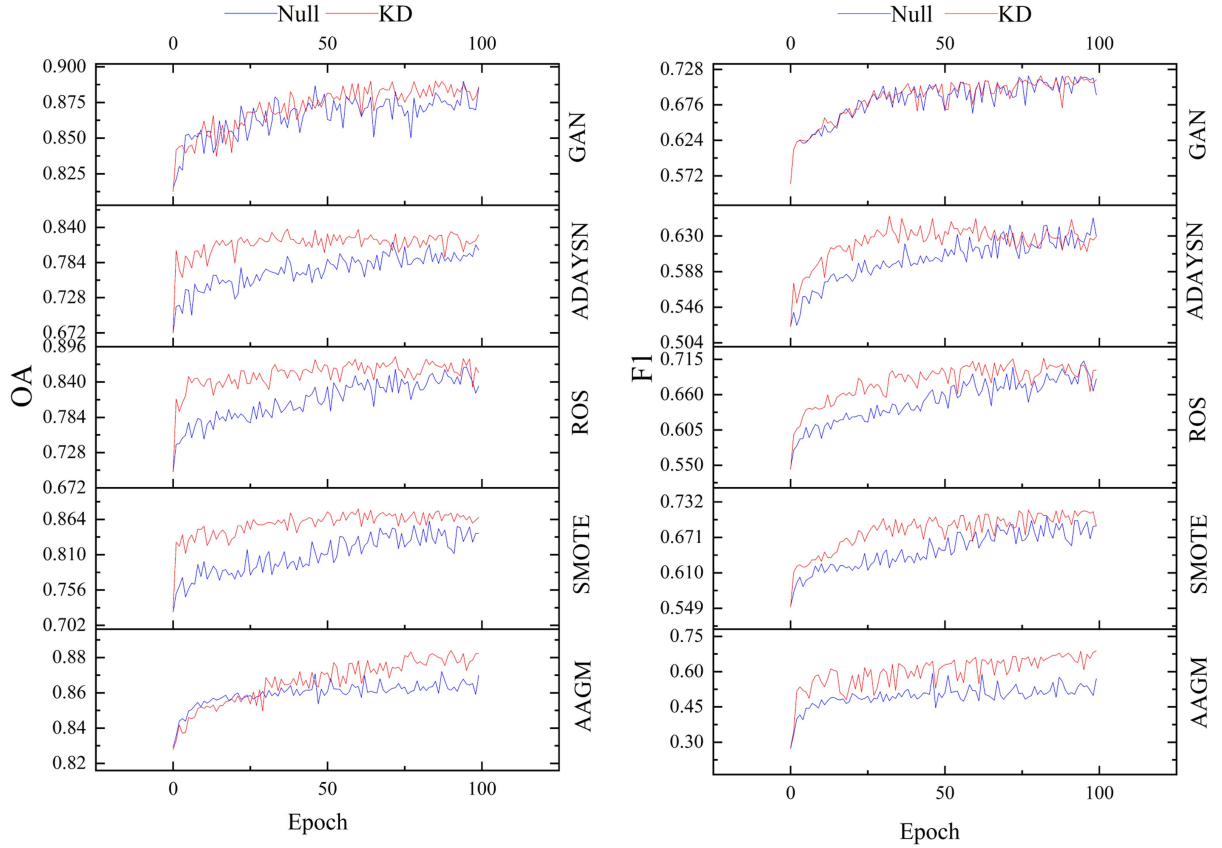


Fig. 5. Comparison line chart of OA and F1 of KD and null on AAGM17.

data augmentation algorithms. During the training process, the classifier trained on the GAN-enhanced data subset serves as the teacher model more frequently. By comparing the curves of each classifier, it is evident that the KD curve consistently outperforms the Null curve, demonstrating the guiding role of CKD. This method successfully transfers the implicit knowledge from the strong classifier to the weak classifier, thereby improving the weak classifier's performance.

From the results in Fig. 6, it can be observed that on the CICIDS17 dataset, each CNN model exhibits distinct advantages and disadvantages in different rounds, with no fixed or frequent CNN consistently serving as the teacher model. All curves utilizing KD outperform those without it. These results indicate that KD plays an effective guiding role, particularly through CKD, where knowledge from high-performing models is successfully transferred to lower-performing models. The CKD strategy, which does not rely on a fixed teacher model, can flexibly select the most suitable teacher model for each task and continuously optimize the student model across multiple training rounds. This approach significantly enhances the model's generalization ability, demonstrating the effectiveness of the CKD method proposed in this chapter.

C. Compare With Other Methods

To validate the effectiveness of the proposed method, it is compared with other existing methods. The comparison algorithms selected are binary classification algorithms

for intrusion detection based on deep learning, including MAGNETO [17], CNN-LSTM [20], HAE-HRL [21], DeepPacket [22], TSIDS [23], RFG-HELAD [7], AlexNet [24], VGG99 [25], MobileNetV2 [26], and ShuffleNet [27]. The comparison experiment uses the entire training set for model training and evaluates the results on the test set. The KDDCUP and UNSW-NB15 datasets are predivided into training and test sets. However, the CICIDS17 and AAGM17 datasets do not provide such divisions. Therefore, this study extracts training and test sets from the original datasets according to the proportions and quantities specified by the comparison algorithms. The final results are presented in Tables III and IV, with the highest evaluation metrics for each dataset highlighted in bold.

As shown in Tables III and IV, KDEHDA outperforms the comparison algorithms on the KDDCUP99, UNSW-NB15, and CICIDS17 datasets. On the AAGM17 dataset, the OA of KDEHDA is 0.8989, slightly lower than MAGNETO's 0.9016. Similarly, the F1-score of KDEHDA reaches 0.7343, while MAGNETO achieves 0.7361. These results indicate a marginal performance gap between the two models on this particular dataset. Unlike the other datasets used in our experiments—which include multiple categories of attack types—the AAGM17 dataset focuses solely on Android malware detection and contains only two types of instances. In such binary classification settings with limited label diversity, the advantages of heterogeneous data augmentation and knowledge ensemble strategies may not be fully exploited. On the contrary, aggressive augmentation can potentially lead to

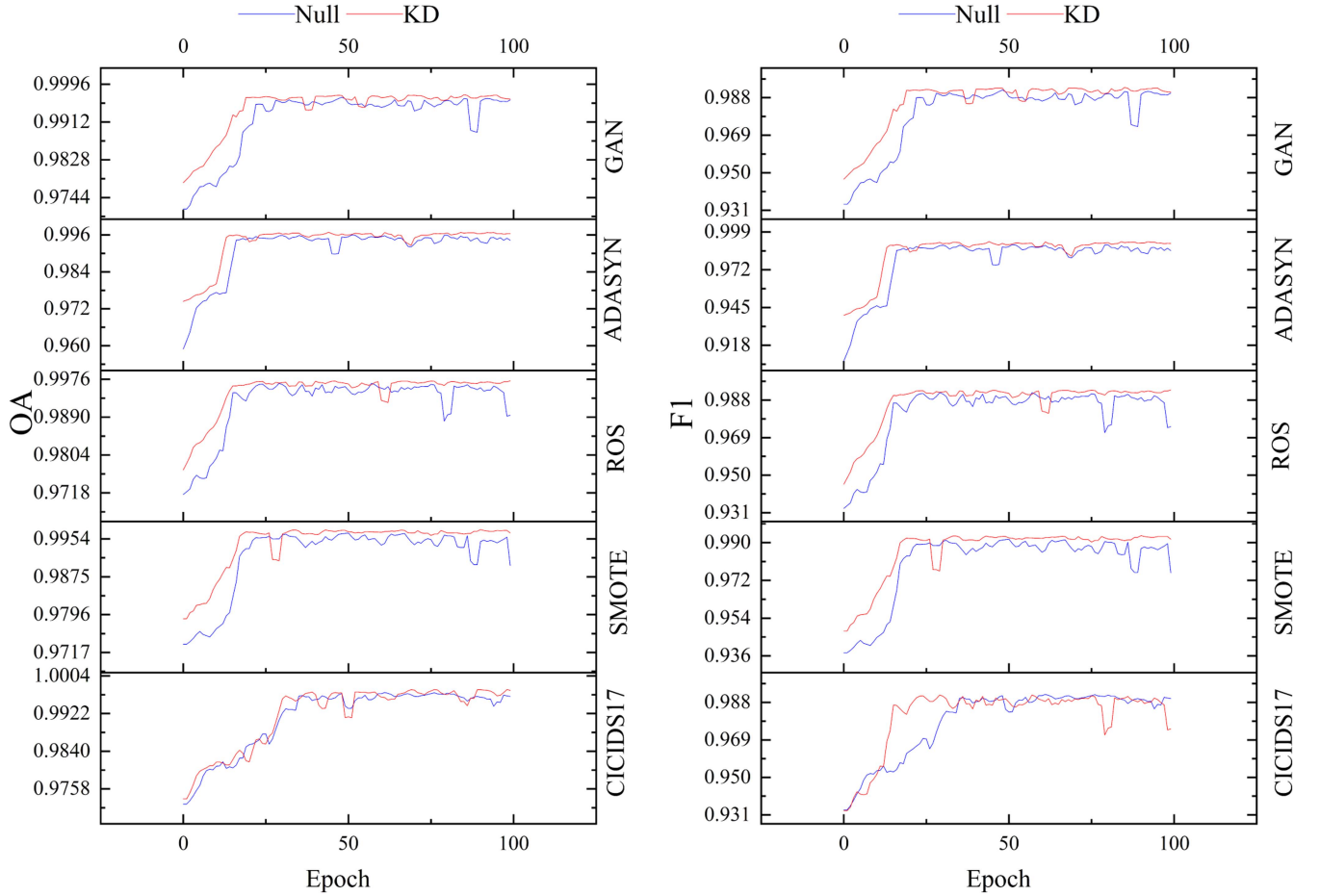


Fig. 6. Comparison line chart of OA and F1 of KD and null on CICIDS17.

TABLE III
OA OF DIFFERENT MODELS ON VARIOUS DATASETS

Dataset	MAGNETO	CNN-LSTM	HAE-HRL	Deep-Packet	TSIDS	RFG-HELAD	AlexNet	VGG99	MobileNetV2	ShuffleNet	KDEHDA
KDDCUP99	0.9329	0.9251	0.9273	0.9256	0.9255	0.9246	0.9258	0.9249	0.9254	0.9243	0.9379
UNSW-NB15	0.8973	0.8939	0.8902	0.8895	0.8937	0.8988	0.9007	0.8489	0.8933	0.883	0.9354
CICIDS17	0.9948	0.9583	0.9781	0.9809	0.9761	0.9856	0.9730	0.9757	0.9694	0.9712	0.9971
AAGM17	0.9016	0.7834	0.8042	0.8284	0.8799	0.8758	0.8861	0.8641	0.8872	0.8897	0.8989
WEB-IDS23	0.9782	0.9756	0.9569	0.9669	0.9909	0.9897	0.9867	0.9763	0.9764	0.9157	0.9932
AVG	0.9410	0.9073	0.9113	0.9183	0.9332	0.9349	0.9345	0.9180	0.9303	0.9168	0.9525

The final results are presented with the highest evaluation metrics for each dataset highlighted in bold.

TABLE IV
F1 OF DIFFERENT MODELS ON VARIOUS DATASETS

Dataset	MAGNETO	CNN-LSTM	HAE-HRL	Deep-Packet	TSIDS	RFG-HELAD	AlexNet	VGG99	MobileNetV2	ShuffleNet	KDEHDA
KDDCUP99	0.9566	0.9513	0.9528	0.9518	0.9517	0.9511	0.9519	0.9513	0.9516	0.9509	0.9603
UNSW-NB15	0.9197	0.9161	0.9137	0.9129	0.9164	0.9121	0.9225	0.8764	0.9157	0.9075	0.9538
CICIDS17	0.9871	0.9733	0.9822	0.9880	0.9849	0.9909	0.9354	0.9359	0.9274	0.9296	0.9929
AAGM17	0.7361	0.6229	0.6374	0.6373	0.6972	0.7251	0.6867	0.6435	0.6757	0.6938	0.7343
WEB-IDS23	0.9476	0.9406	0.8817	0.9525	0.9709	0.9746	0.9676	0.9441	0.9442	0.8199	0.9832
AVG	0.9094	0.8808	0.8736	0.8885	0.9042	0.9108	0.8928	0.8702	0.8829	0.8603	0.9249

The final results are presented with the highest evaluation metrics for each dataset highlighted in bold.

overfitting due to the lack of interclass variability. Therefore, the slightly inferior performance of KDEHDA on AAGM17 is likely due to the reduced effectiveness.

The superior performance of KDEHDA, compared to other algorithms, can be attributed to its use of multiple heterogeneous

integration methods. Intrusion detection tasks typically involve processing tabular data, which differs from image data in terms of processing techniques. Traditional image data enhancement techniques, such as cropping and inversion, are designed for image recognition models, such as AlexNet and are not fully

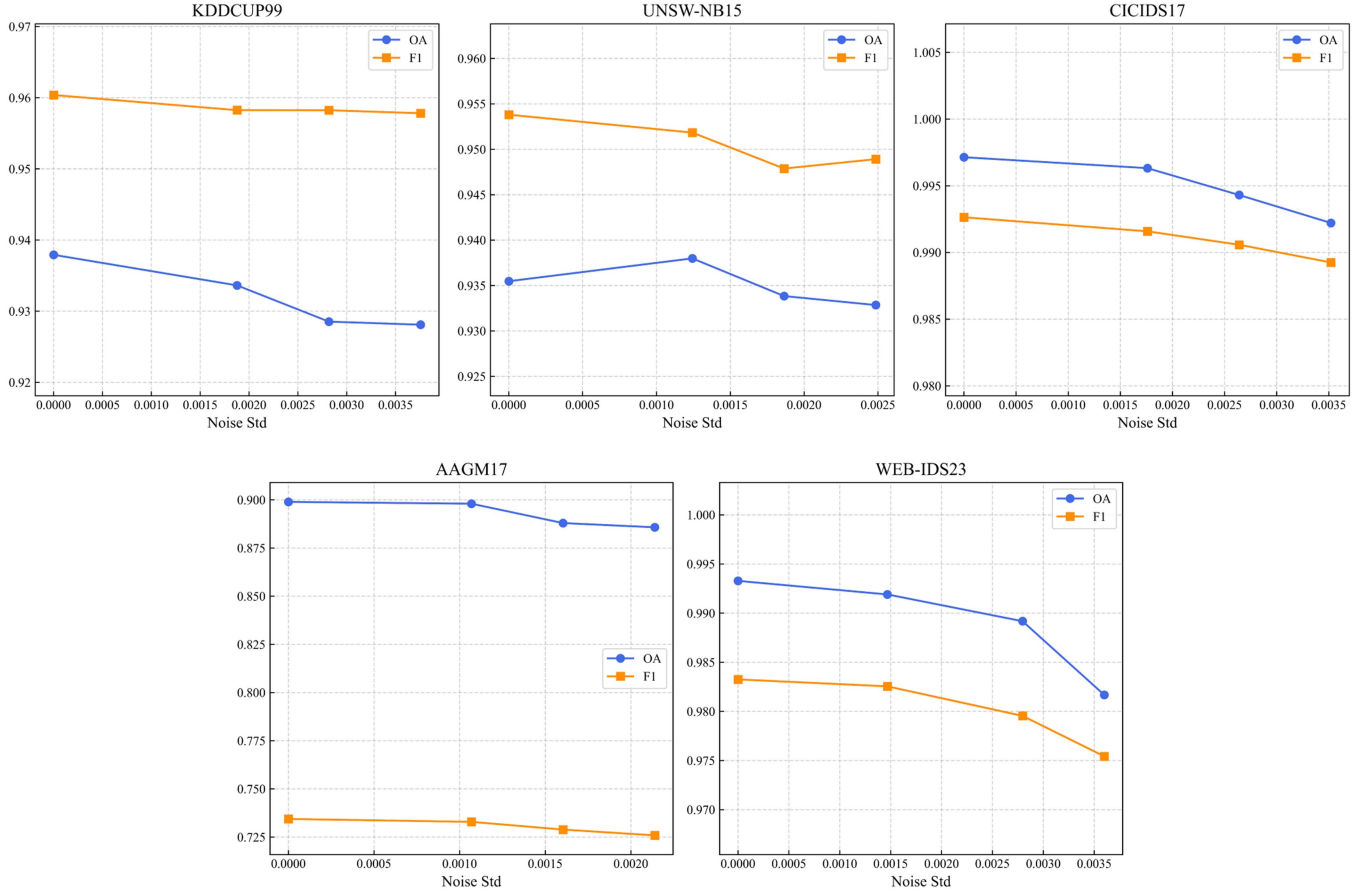


Fig. 7. Robustness experiment of KDEHDA under Gaussian noise with varying σ values.

applicable to tabular data. In contrast, MAGNETO relies solely on GAN to address data imbalance. Although simple and effective, this approach may not fully capture the complex features in the data. KDEHDA enhances the model's generalization and robustness by utilizing multiple heterogeneous integration methods. This approach not only enhances the dataset, but also enables multiple models to learn from each other through KD, further boosting detection performance. KDEHDA combines the advantages of different models through heterogeneous integration, enhancing adaptability to data diversity. In addition, KD facilitates mutual learning and knowledge transfer between models, playing a key role in improving classification accuracy and preventing overfitting. One significant advantage of KDEHDA over other KD methods is that it does not require a pretrained teacher model. This feature significantly reduces training costs and time overhead. Traditional KD methods typically require pretraining a teacher model and transferring its knowledge to the student model, increasing both training complexity and resource demands. KDEHDA omits this step through distillation integration, making the training process more efficient and cost-effective.

To further illustrate the efficiency and practicality of the proposed KDEHDA model, we conduct a quantitative analysis of model complexity and computational cost. As shown in Table V, KDEHDA has significantly fewer parameters (0.1939 M) and lower FLOPs (28.34 M) compared to most

TABLE V
MODEL PARAMETERS AND FLOPS

Model	#Params (M)	#FLOPs (M)
MAGNETO	3.26	167.14
CNN-LSTM	0.44	23.46
HAE-HRL	0.0916	157.239
Deep-Packet	3.70	61.70
RFG-HELAD	2.03	98
TSIDS	1.32	254.80
AlexNet	61.00	724.00
VGG99	143.00	19600.00
MobileNetV2	3.50	300.00
ShuffleNet	1.00	140.00
KDEHDA	0.1939	28.34

baseline models, including heavyweight architectures, such as AlexNet (61.00 M parameters) and VGG99 (143.00 M parameters). Even when compared with lightweight models such as MobileNetV2 (3.50 M parameters) and ShuffleNet (1.00 M parameters), KDEHDA remains more compact and efficient. This result supports the claim that KDEHDA is not only accurate, but also highly resource-efficient, making it well-suited for deployment in real-time or resource-constrained environments.

D. Robustness Experiment Under Gaussian Noise

To evaluate the robustness of the proposed KDEHDA model under real-world noisy conditions, Gaussian white noise with

zero mean and varying variance σ^2 was added to the input features during the testing phase. The perturbed samples $\tilde{\mathbf{x}} = \mathbf{x} + \varepsilon$ simulate potential sensor noise or environmental disturbances that may occur during model deployment. The training data remained unchanged to isolate the model's generalization capability under noise.

The experimental results are illustrated in Fig. 7, which shows the variation in OA and F1-score with increasing noise intensity σ . As expected, both OA and F1-score exhibit a gradual decline as σ increases. However, the degradation remains relatively minor—less than 0.01—across all datasets, indicating strong noise resilience.

These findings confirm that the proposed KDEHDA model not only achieves high classification performance under clean conditions, but also maintains robust and stable behavior under noisy perturbations. This robustness makes it well-suited for practical deployment in real-world intrusion detection scenarios.

V. CONCLUSION

Intrusion detection data typically exhibits significant class imbalance, with normal traffic instances far outnumbering attack traffic instances [28]. Accurately identifying attack traffic is crucial in practical applications, as it directly impacts the system's security and protection capabilities. We propose an innovative heterogeneous data augmentation ensemble learning framework that improves dataset diversity by combining multiple augmentation algorithms, thereby enhancing the model's ability to identify minority classes. In addition, we introduce CKD. This method transfers data diversity through the base model during training, further enhancing the model's generalization ability. CKD transfers knowledge between models, making the model more comprehensive and deeper in learning data features. This enables the model to exhibit greater robustness and accuracy when handling new instances. We use multiple shallow convolutional neural networks as base classifiers in our approach. This approach significantly reduces model complexity and computational overhead, while maintaining high accuracy.

In summary, our solution, combining heterogeneous data-enhanced ensemble learning and convolution kernel KD, improves intrusion detection model performance, while ensuring an efficient training process, offering broad practical application potential. Nonetheless, our method also has some limitations. First, the use of multiple data augmentation algorithms and knowledge transfer mechanisms increases the complexity of the training pipeline, which may lead to higher preprocessing and training time. Second, the effectiveness of our approach may depend on the availability and quality of labeled intrusion datasets, especially for rare attack types. In future work, we plan to explore automatic selection of augmentation strategies and extend our method to unsupervised or semi-supervised scenarios to alleviate data dependency.

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